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Prove F test is equal to T test squared

Asked 11 years, 2 months ago Modified 5 years, 2 months ago Viewed 65k times

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I need to show that F test is equal to T test squared, when the T test is for 2 independent groups and assuming variances are equal.

I know that $F = \frac{MSB}{MSW} = \frac{SSB/k-1}{SSW/N-K}$ and I know that $T = \frac{X-Y}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}}$,

so $T^2 = \frac{(X-Y)^2}{S_p^2(\frac{1}{n}+\frac{1}{m})}$

I've seen this proof in Regression but here we're not using MSE and MSR, so i'm not sure how to connect between the two.

self-study mathematical-statistics anova t-test f-test

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edited Nov 4, 2018 at 8:19
Ferdi
5,179 9 46 64

asked Apr 5, 2013 at 12:17
Manko
521 2 5 12

Any proof is a series of steps that logically connect hypotheses to a conclusion. Although your conclusion is clearly stated, what hypotheses do you want to begin with? After all, because both procedures test the same thing by means of the same hypothesis-testing framework using the same assumptions, on that account alone they *must* be equivalent! So if that's not a sufficient proof for you, please indicate where you are beginning and what methods of proof are desired. – whuber ♦ Apr 5, 2013 at 13:20

3 Answers

Sorted by: Highest score (default)

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I have done this proof in my blog
Since I already have the code for the equations, I'm reproducing it here.

We have to prove that

$$F_{a-1,N-a} = \frac{MST}{MSE} = \frac{\frac{SST}{a-1}}{\frac{SSE}{N-a}} \tag{1}$$

reduces to

$$t_k^2 = \frac{(\bar{y}_{1.} - \bar{y}_{2.})^2}{S_p^2(\frac{1}{n_1} + \frac{1}{n_2})} \tag{2}$$

$\mathrm{When\ a} = 2$ (this is key)

Notation

SSE: Sum of Squares due to Error
SST: Sum of Squares of Treatment
MSE: Mean Sum of squares Error
MST: Mean Sum of squares Treatment
a: Number of treatments
n_1: Number of observations in treatment 1
n_2: Number of observations in treatment 2
N: Total number of observations
$\bar{y}_{i.}$: Mean of treatment i
$\bar{y}_{..}$: Global mean
$k = N - a$: Degrees of freedom of the denominator of F

Now that we have the formulas, we will work the following:

1. Denominator of equation (1)

1. Denominator of equation (1)

When $a = 2$ the denominator of expression (1) is:

$$MSE = \frac{SSE}{N-2} = \frac{\sum_{j=1}^{n_1}(y_{1j} - \bar{y}_{1.})^2 + \sum_{j=1}^{n_2}(y_{2j} - \bar{y}_{2.})^2}{N-2} \tag{3}$$

Recalling that the formula for the sample variance estimator is,

$$S_i^2 = \frac{\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i.})^2}{n_i - 1}$$

we can multiply and divide the terms in the numerator in (3) by $(n_i - 1)$ and get (4). Don't forget that in this case $N = n_1 + n_2$

$$\frac{SSE}{N-2} = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2} = S_p^2 \tag{4}$$

S_p^2 is called the *pooled variance estimator*.

2. Numerator of equation (1)

When $a = 2$ the numerator of expression (1) is:

$$\frac{SST}{2-1} = SST$$

and the general expression for SST reduces to $SST = \sum_1^2 n_i(\bar{y}_{i.} - \bar{y}_{..})^2$. The next step is to expand the sum as follows:

$$SST = \sum_1^2 n_i(\bar{y}_{i.} - \bar{y}_{..})^2 = n_1(\bar{y}_{1.} - \bar{y}_{..})^2 + n_2(\bar{y}_{2.} - \bar{y}_{..})^2 \tag{5}$$

$\bar{y}_{..}$ is called the *global mean* and we are going to write it in a different way. The new way is:

$$\bar{y}_{..} = \frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N} \tag{6}$$

Next, replace (6) in formula (5) and re-write SST as:

$$SST = \underbrace{n_1\big[\bar{y}_{1.} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2}_{\text{Part a}} + \underbrace{n_2\big[\bar{y}_{2.} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2}_{\text{Part b}} \tag{7}$$

The next step is to find alternative ways for the expressions **Part a** and **Part b**

2.a. Part a

$$\text{Part a} = n_1\big[\bar{y}_{1.} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2$$

Multiply and divide the term with $\bar{y}_{1.}$ by N

$$n_1\big[\frac{N\bar{y}_{1.}}{N} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2$$

N is common denominator

$$n_1\big[\frac{N\bar{y}_{1.} - n_1\bar{y}_{1.} - n_2\bar{y}_{2.}}{N}\big]^2$$

$\bar{y}_{1.}$ is common factor of N and n_1

$$n_1\big[\frac{(N - n_1)\bar{y}_{1.} - n_2\bar{y}_{2.}}{N}\big]^2$$

Replace $(N - n_1) = n_2$

$$n_1 \big[\frac{n_2\bar{y}_{1.} - n_2\bar{y}_{2.}}{N}\big]^2$$

Now n_2 is common factor of $\bar{y}_{1.}$ and $\bar{y}_{2.}$

$$n_1 \big[\frac{n_2(\bar{y}_{1.} - \bar{y}_{2.})}{N}\big]^2$$

Take n_2 and N out of the square

$$\text{Part a} = \frac{n_1 n_2^2}{N^2}(\bar{y}_{1.} - \bar{y}_{2.})^2$$

2.b. Part b

$$\text{Part b} = n_2 \big[\bar{y}_{2.} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2$$

Multiply and divide the term with $\bar{y}_{2.}$ by N

$$n_2 \big[\frac{N\bar{y}_{2.}}{N} - (\frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.}}{N})\big]^2$$

N is common denominator

$$n_2 \big[\frac{N\bar{y}_{2.} - n_1\bar{y}_{1.} - n_2\bar{y}_{2.}}{N}\big]^2$$

$\bar{y}_{2.}$ is common factor of N and n_2

$$n_2 \big[\frac{(N - n_2)\bar{y}_{2.} - n_1\bar{y}_{1.}}{N}\big]^2$$

Replace $(N - n_2) = n_1$

$$n_2 \big[\frac{n_1\bar{y}_{2.} - n_1\bar{y}_{1.}}{N}\big]^2$$

Now n_1 is common factor of $\bar{y}_{1.}$ and $\bar{y}_{2.}$

$$n_2 \big[\frac{n_1(\bar{y}_{2.} - \bar{y}_{1.})}{N}\big]^2$$

Take n_1 and N out of the square

$$\text{Part b} = \frac{n_2 n_1^2}{N^2}(\bar{y}_{2.} - \bar{y}_{1.})^2$$

Now that we have **Part a** and **Part b** we are going to go back to equation (7) and replace them:

$$SST = \frac{n_1 n_2^2}{N^2}(\bar{y}_{1.} - \bar{y}_{2.})^2 + \frac{n_2 n_1^2}{N^2}(\bar{y}_{2.} - \bar{y}_{1.})^2 \tag{8}$$

Taking into account that $(\bar{y}_{1.} - \bar{y}_{2.})^2 = (\bar{y}_{2.} - \bar{y}_{1.})^2$, we can re-write equation (8) as (9):

$$SST = \underbrace{\big[\frac{n_1 n_2^2}{N^2} + \frac{n_2 n_1^2}{N^2}\big]}_{\text{Part c}}(\bar{y}_{1.} - \bar{y}_{2.})^2 \tag{9}$$

This lead us with part **Part c**, that we are going to work next.

2.c. Part c

$$\text{Part c} = \frac{n_1 n_2^2}{N^2} + \frac{n_2 n_1^2}{N^2}$$

N^2 is common denominator and each of the summands has a $n_1 n_2$ factor that we can factor out. Then we have:

$$\frac{n_1 n_2(n_1 + n_2)}{N^2}$$

Replace $N = n_1 + n_2$

$$\frac{n_1 n_2 N}{N^2}$$

Simplify N

$$\frac{n_1 n_2}{N}$$

Re-write the fraction

$$\frac{1}{\frac{N}{n_1 n_2}}$$

Replace $N = n_1 + n_2$

$$\frac{1}{\frac{n_1+n_2}{n_1 n_2}} = \frac{1}{\frac{1}{n_1} + \frac{1}{n_2}}$$

And we have

$$\text{Part c} = \frac{1}{\frac{1}{n_1} + \frac{1}{n_2}}$$

Finally, we have to replace this expression for **Part c** in (9) and re-write SST as:

$$SST = \frac{1}{\frac{1}{n_1} + \frac{1}{n_2}}(\bar{y}_{1.} - \bar{y}_{2.})^2$$

3. Put all together

With the previous steps we have shown that, $\color{red}{\text{when a} = 2}$, we have:

$$\frac{SST}{2-1} = \frac{(\bar{y}_{1.} - \bar{y}_{2.})^2}{\frac{1}{n_1} + \frac{1}{n_2}}$$

and

$$\frac{SSE}{N-2} = S_p^2$$

The ratio of these two expressions, namely the F-statistic, is then:

$$F_{1,k} = \frac{\frac{SST}{2-1}}{\frac{SSE}{N-2}} = \frac{(\bar{y}_{1.} - \bar{y}_{2.})^2}{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = t_k^2$$

And this concludes the proof.

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edited Apr 11, 2019 at 15:09

answered Nov 4, 2018 at 0:56

canovasjm
383 1 3 6

8

Because one has $\boxed{T^2 = F}$.

To show that, you have to check that (with $N = mn$):

- $SSW/(N-2) = S_p^2$ (the unbiaised estimate of σ^2)
- $SSB = (\bar{X} - \bar{Y})^2/(\frac{1}{n} + \frac{1}{m})$

To show the second point you only have to use :

- the elementary equality $SSB = m(\bar{x} - \bar{x \cdot y})^2 + n(\bar{y} - \bar{x \cdot y})^2$
- the fact that the mean of the whole sample $x \cdot y = (x_1, \ldots, x_m, y_1, \ldots y_n)$ is the weighted mean $\frac{m\bar{x}+n\bar{y}}{m+n}$
- some elementary but a little tiedous calculations to conclude

Sorry for the strange notation $x \cdot y$ for the "whole sample", this was my first idea and I'm in a hurry now.

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answered Apr 5, 2013 at 19:55

Stéphane Laurent
19.4k 5 74 109

Stéphane, can we then say that their power is equal? – An old man in the sea. May 8, 2017 at 19:35

@Anoldmaninthesea.yes, these are the same tests. – Stéphane Laurent May 8, 2017 at 22:04

So, we can say that about a t-test and an F-test when we're testing $H_0 : \beta_i = 0$ vs $H_1 : \beta_i \neq 0$, right? – An old man in the sea. May 9, 2017 at 7:31

4

You can rewrite the equation as

$$\frac{SSB/\left(k-1\right)}{SSW/\left(N-k\right)} = \frac{SSB\left(k-1\right)/\sigma^2(k-1)^2}{SSW\left(N-k\right)/\sigma^2(N-k)^2}$$

For $k = 2$ (two groups),

$$\frac{SSB\left(k-1\right)/\sigma^2(k-1)^2}{SSW\left(N-k\right)/\sigma^2(N-k)^2} = \frac{SSB/\sigma^2}{SSW\left(N-2\right)/\sigma^2(N-2)^2}.$$

The numerator is a χ^2 distribution with one degree of freedom. The denominator has the following distribution:

$$SSW\left(N-2\right)/\sigma^2(N-2)^2 \sim \frac{\chi^2_{N-2}}{(N-2)^2}.$$

Therefore, you have the ratio of two χ^2 distributions. This ratio is equivalent to a t distribution with $N-1$ degrees of freedom squared:

$$\frac{\chi^2_1}{\chi^2_{N-2}/(N-2)^2} \sim t^2_{N-2}.$$

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edited Apr 5, 2013 at 20:10

answered Apr 5, 2013 at 13:59
wcampbell
2,197 1 18 20

Why are we assuming that k=1? And regarding the equality, shouldn't the numerator 0 if k=1 because k-1=0? – Manko Apr 5, 2013 at 15:50

Sorry, k = 2 because you have two groups. I edited my post. – wcampbell Apr 5, 2013 at 15:54

:) so shouldn't $SSW(N-1)/\sigma 2(N-1)^2$ be $SSW(N-2)/\sigma 2(N-2)^2$..? and we when reach the end of the equation.. we reach t_{N-1}? – Manko Apr 5, 2013 at 15:57

Another mistake! I fixed that one too. At the end, you get t_{N-2} which is the correct number of degrees of freedom for a two sample t test. – wcampbell Apr 5, 2013 at 17:47

5 You don't prove that $F = T^2$, you only prove that F has *the same law* as T^2. By the way you make a mistake: $\sqrt{F}$ has not a t-distribution since it is distributed on positive numbers. – Stéphane Laurent Apr 5, 2013 at 19:58